PROBLEM 17

Linear Central Force

Original statement · Problems.pdf, p. 9

Newtonian Restoring Force

A force attracts a heavy particle M of mass m to a point O. The attractive force is proportional to the mass of the particle and to the distance OM. The proportionality coefficient is k^2. The coordinate system $Oxyz$ has the origin at the centre of attraction O and the axis Oz is vertically oriented. Initially, the coordinates of M are:

$$(x_0, y_0, z_0) = \left(\frac{\sqrt{3}g}{k^2}, 0, 0\right)$$

and its initial velocity is:

$$(\dot{x}_0, \dot{y}_0, \dot{z}_0) = \left(0, \frac{2g}{k}, 0\right)$$

(a) Describe the motion of M, specifying clearly the shape of the path.

(b) What is the velocity of the particle as a function of t?

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a heavy particle M of mass m is attracted to the fixed point O by a force proportional to its mass and to the distance, with coefficient k^2: $\mathbf{F}_c = -mk^2\mathbf{r}$. The axis Oz is vertical (upward) and gravity acts along $-Oz$.
- **Constraints:** none; the particle is free in space, so $\mathrm{CDOF} = 3$, with the Cartesian coordinates (x, y, z).
- **Initial state:** $(x_0, y_0, z_0) = \left(\sqrt{3}g/k^2,\ 0,\ 0\right)$ and $(\dot{x}_0, \dot{y}_0, \dot{z}_0) = (0,\ 2g/k,\ 0)$.
- **Parameters:** m [kg], g [m/s^2], k [s^{-1}].

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with its origin at the centre of attraction and $\mathbf{k}$ vertical upward (constant unit vectors).

Derivation

3.1 Kinematics

$$\mathbf{r} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}, \qquad \mathbf{v} = \dot{x}\,\mathbf{i} + \dot{y}\,\mathbf{j} + \dot{z}\,\mathbf{k}, \qquad \mathbf{a} = \ddot{x}\,\mathbf{i} + \ddot{y}\,\mathbf{j} + \ddot{z}\,\mathbf{k}$$

3.2 Forces

- **Applied forces:** the linear central attraction $\mathbf{F}_c = -mk^2\mathbf{r}$ and the weight $\mathbf{F}_g = -mg\,\mathbf{k}$. **Constraint forces:** none. **Inertia forces:** none.

3.3 Equations of Motion

- **Newton's second law in S_0:** $m\ddot{\mathbf{r}} = -mk^2\mathbf{r} - mg\,\mathbf{k}$.

> **Pedagogical Justification:** the system is linear and the basis is Cartesian and constant, so the vector equation decouples into three independent scalar equations with constant coefficients: three unknowns, three equations.

$$\ddot{x} + k^2 x = 0, \qquad \ddot{y} + k^2 y = 0, \qquad \ddot{z} + k^2 z = -g$$

3.4 Integration and Answers

Integration of each component

- x **and** y **(homogeneous oscillators):** the equation $\ddot{x} + k^2 x = 0$ is linear with constant coefficients and its characteristic equation $r^2 + k^2 = 0$ has the roots $\pm ik$ (the characteristic-equation method of Notes Eq. 5.20, Sec. 5.3), so the general solution is $A\cos kt + B\sin kt$ (Notes Sec. 5.1, Eq. 5.7). Its velocity follows from the chain rule, $\frac{d}{dt}\cos kt = -k\sin kt$ and $\frac{d}{dt}\sin kt = k\cos kt$: $\dot{x} = -kA\sin kt + kB\cos kt$. The initial conditions give

$$x(0) = A_x = \frac{\sqrt{3}g}{k^2}, \;\; \dot{x}(0) = kB_x = 0 \implies x(t) = \frac{\sqrt{3}g}{k^2}\cos kt$$

$$y(0) = A_y = 0, \;\; \dot{y}(0) = kB_y = \frac{2g}{k} \implies B_y = \frac{2g}{k^2} \implies y(t) = \frac{2g}{k^2}\sin kt$$

- z **(forced by gravity):** a particular solution of $\ddot{z} + k^2 z = -g$ is the constant $z_p = -g/k^2$, so $z(t) = -\dfrac{g}{k^2} + A_z\cos kt + B_z\sin kt$. The initial conditions give $z(0) = -\dfrac{g}{k^2} + A_z = 0$, so $A_z = \dfrac{g}{k^2}$, and $\dot{z}(0) = kB_z = 0$:

$$z(t) = \frac{g}{k^2}\left(\cos kt - 1\right)$$

- **Trajectory:** $\mathbf{r}(t) = \dfrac{\sqrt{3}g}{k^2}\cos kt\,\mathbf{i} + \dfrac{2g}{k^2}\sin kt\,\mathbf{j} + \dfrac{g}{k^2}\left(\cos kt - 1\right)\mathbf{k}$.

(a) Shape of the path

> **Pedagogical Justification:** a linear relation among the coordinates indicates a plane, and a constant distance to a point indicates a sphere; their intersection is a circle.

- **Plane.** From $x = \frac{\sqrt{3}g}{k^2}\cos kt$ we have $\cos kt = \frac{k^2}{\sqrt{3}g}x$, so $z = \frac{g}{k^2}\cos kt - \frac{g}{k^2} = \frac{x}{\sqrt{3}} - \frac{g}{k^2}$, i.e.

$$x - \sqrt{3}\,z = \frac{\sqrt{3}g}{k^2}$$

- **Sphere.** The squared distance to the point $C = \left(0, 0, -g/k^2\right)$ is

$$x^2 + y^2 + \left(z + \frac{g}{k^2}\right)^2 = \frac{3g^2}{k^4}\cos^2 kt + \frac{4g^2}{k^4}\sin^2 kt + \frac{g^2}{k^4}\cos^2 kt = \frac{4g^2}{k^4}\left(\cos^2 kt + \sin^2 kt\right) = \frac{4g^2}{k^4}$$

a sphere of centre C and radius $R = 2g/k^2$.

- **Circle.** The centre C lies on the plane: $0 - \sqrt{3}\left(-g/k^2\right) = \sqrt{3}g/k^2$. The intersection of a sphere with a plane through its centre is a **great circle** of radius $R = 2g/k^2$.

(b) Velocity as a function of time

- Differentiating each component with the chain rule:

$$\dot{x} = \frac{\sqrt{3}g}{k^2}\left(-k\sin kt\right), \qquad \dot{y} = \frac{2g}{k^2}\left(k\cos kt\right), \qquad \dot{z} = \frac{g}{k^2}\left(-k\sin kt\right)$$

$$\mathbf{v}(t) = -\frac{\sqrt{3}g}{k}\sin kt\,\mathbf{i} + \frac{2g}{k}\cos kt\,\mathbf{j} - \frac{g}{k}\sin kt\,\mathbf{k}$$

- **Modulus:** $\|\mathbf{v}\|^2 = \frac{g^2}{k^2}\left[3\sin^2 kt + 4\cos^2 kt + \sin^2 kt\right] = \frac{g^2}{k^2}\cdot 4\left(\sin^2 kt + \cos^2 kt\right) = \frac{4g^2}{k^2}$, so $\|\mathbf{v}\| = \frac{2g}{k}$: the speed is constant.

Interpretation & Checks

- **Uniform circular motion:** the particle describes a great circle of radius $R = 2g/k^2$ at constant speed $v = 2g/k$, with angular frequency $\omega = v/R = k$.
- **Shifted centre:** the linear central force plus the uniform gravity is equivalent to a spring centred at the equilibrium point C where $k^2 z_{\text{eq}} = -g$, i.e. $z_{\text{eq}} = -g/k^2$.
- **Consistency:** the centripetal acceleration $v^2/R = (4g^2/k^2)/(2g/k^2) = 2g$ equals the effective spring acceleration $k^2 R = 2g$.
- **Role of k:** the initial data are themselves multiples of g/k^2 ($x_0 = \sqrt{3}\, g/k^2$, $\dot{y}_0 = 2g/k$), so the solution is self-similar: R, $|z_{\text{eq}}|$ and x_0 all scale with g/k^2, the angular frequency is k, and the geometry (offset over radius $|z_{\text{eq}}|/R = 1/2$, tilt of the plane, ratio $1/2$ of gravity to central force on the orbit) does not depend on k. A larger k only shrinks the same orbit; gravity never becomes negligible for these data.
- **Dimensional check:** $[g/k^2] = \dfrac{\mathrm{m\,s^{-2}}}{\mathrm{s^{-2}}} = \mathrm{m}$, $[g/k] = \mathrm{m/s}$, $[R] = \mathrm{m}$ and $\omega = v/R = k$ in $\mathrm{s^{-1}}$.
- **Check against the official key:** the key of `Problems.pdf` gives $\mathbf{r}_0^M = \frac{\sqrt{3}g}{k^2}\cos kt\,\mathbf{i} + \frac{2g}{k^2}\sin kt\,\mathbf{j} + \frac{g}{k^2}(\cos kt - 1)\,\mathbf{k}$, the circle as the intersection of the sphere $x^2 + y^2 + (z + g/k^2)^2 = 4g^2/k^4$ with the plane $x - \sqrt{3}z = \sqrt{3}g/k^2$, and the constant speed $2g/k$. All coincide with the results above. Sympy confirms $\ddot{\mathbf{r}} + k^2\mathbf{r} + g\mathbf{k} = \mathbf{0}$, the initial data, $\|\mathbf{v}\|^2 = 4g^2/k^2$, the plane and the sphere with zero residual.

Source: Problems.pdf